

PHASE 1 – IDEATION PHASE

1.1 Project Overview

The **Medical Inventory Management System** is a Salesforce-based cloud application designed to streamline and automate the process of managing medical supplies, medicines, and related healthcare items.

In most hospitals and medical centers, inventory management is done manually using spreadsheets or traditional systems, which leads to inefficiency and human error.

This project leverages the **Salesforce CRM platform** to provide an integrated solution where users can manage inventory data, track medicine batches, monitor stock levels, and receive real-time alerts for low or expired items.

The system also provides dashboards and analytical reports that help healthcare administrators make informed decisions regarding stock purchases, supplier performance, and medicine usage patterns.

Ultimately, the solution aims to improve **efficiency, accuracy, and patient care** by ensuring that essential medicines are always available.

1.2 Objectives

The main objective of this project is to build a scalable and efficient **Salesforce application** for managing medical inventory.

Specific goals include:

1. **Automation:** Eliminate manual processes by automating stock updates, purchase approvals, and expiry alerts using Salesforce Flows and Apex Triggers.
 2. **Accuracy:** Maintain real-time and error-free data regarding medicines, suppliers, and stock levels.
 3. **Transparency:** Provide a centralized system for tracking all inventory-related activities.
 4. **Reporting:** Generate detailed analytical reports and dashboards to monitor stock usage and forecast demand.
 5. **Security:** Ensure data protection through Salesforce's robust role-based access control.
 6. **Scalability:** Allow the system to adapt to hospitals or clinics of different sizes and requirements.
 7. **Integration:** Support possible integration with external systems like billing or patient management software in the future.
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1.3 Problem Statement

The medical field relies heavily on having the right medicines available at the right time. However, many hospitals and pharmacies face challenges such as:

- **Manual tracking of stock levels**, leading to errors or delays in reordering.
- **Lack of real-time visibility** into medicine availability and expiry dates.
- **Wastage due to expired medicines** and overstocking.
- **Difficulty in supplier coordination** and purchase order tracking.
- **Time-consuming reporting** and stock verification processes.

These issues affect the **efficiency, accuracy, and reliability** of healthcare operations. Therefore, there is a critical need for a **digital inventory management solution** that minimizes manual work, reduces costs, and ensures timely availability of medical items.

1.4 Proposed Solution

To address the above challenges, the **Medical Inventory Management System** will be built on the **Salesforce Lightning Platform** to provide an end-to-end automated solution for inventory tracking and management.

Key proposed features include:

1. Custom Salesforce Objects:

- Medicine
- Supplier
- Purchase Order
- Inventory Transaction

Each object will store relevant data and maintain relationships to ensure accurate stock tracking.

2. Automation through Salesforce Flow and Apex:

- Automatic update of stock levels after purchase or issue.
- Email alerts and notifications for low stock or expiring medicines.
- Automated approval process for purchase orders.

3. Reports and Dashboards:

- Real-time dashboards showing total stock, expiring medicines, and supplier performance.
- Custom reports for management review and audit tracking.

4. User Roles and Permissions:

- Admin, Manager, and Pharmacist roles with specific permissions.
- Secure data sharing using Salesforce Profiles and Permission Sets.

5. Scalability and Cloud Advantage:

- Accessible from any device via Salesforce cloud.
- Easily extendable with other Salesforce apps or third-party integrations.

6. Benefits:

- Reduces manual effort and errors.
 - Enhances efficiency and transparency.
 - Improves planning and decision-making.
 - Ensures compliance and accountability in stock handling.
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1.5 Expected Outcome

By implementing this Salesforce-based Medical Inventory Management System, healthcare organizations can achieve:

- Efficient inventory control and management.
- Reduced medicine wastage and shortages.
- Improved accuracy in stock reporting.
- Enhanced user satisfaction and operational productivity.
- A modern, scalable, and reliable cloud solution aligned with healthcare needs.